

A Working Guide to the Wrong- θ Critique of Conventional Intra-National Distance

Volume 2

Numerical Evaluation, Empirics, and the Forward Path

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Companion to Volume 1 (Trade Economics Foundations, the Structural CES Gravity Model, Internal Distance, Generalized Means and the Divergence). This volume picks up where Volume 1 ends and develops the numerical, empirical, and methodological apparatus needed to take the wrong- θ critique into estimation practice.

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Preface to Volume 2

Volume 1 ended on a structural observation: the Head-Mayer (2002) closed-form expression for intra-national distance has a pole at $\theta = -2$, the underlying integral diverges in 2D for $\theta \leq -2$, and the empirically defensible $\theta = \delta(1 - \sigma) \in [-7, -3]$ for $\sigma \in [4, 8]$ falls entirely below the pole. Volume 2 takes that observation into estimation practice.

The volume has five parts.

Part V (Numerical Evaluation and Regularization) develops the numerical apparatus. If the integral diverges as a continuous object, the analyst must choose a finite spatial grain $d_{\min}$ at which to evaluate the discretized version. The implied $d_{ii}^{\text{eff}}(\theta, d_{\min})$ varies sharply with that choice. Part V documents the regularization curve on a uniform disk (the cleanest case), then on real country polygons via the Mollweide projection.

Part VI (Real-Data Construction) walks through the GIS pipeline. BACI for bilateral trade flows. CEPII Gravity for size, language, contiguity, and the canonical intra-national distance. GHS-POP, WorldPop, and VIIRS for population and economic-activity rasters. OpenStreetMap for transport-network distance. The empirical machinery behind the empirical claim.

Part VII (Regression Specification) is the methodological core. OLS gravity with origin and destination fixed effects. The perfect-collinearity trap that corrupts coefficients when log-GDP regressors are included alongside country FE. The Poisson PML alternative. The Yotov-Piermartini-Monteiro-Larch pair-year FE convention. A working analyst should leave Part VII with the ability to estimate a clean gravity regression from raw data.

Part VIII (Empirical Results) walks through the BACI-CEPII regularization sweep. The Spec A baseline, the Spec B-prime sweep, the sign-flip threshold by structural θ , the cross-year stability problem. The interpretation: a coefficient that the literature has treated as a structural border-effect parameter is in fact a function of an analyst-chosen discretization grain.

Part IX (Forward Path) sketches what an honest gravity literature would do. The Anderson-Yotov multilateral-resistance system over time. Sectoral and pair-year FE specifications. Welfare counterfactuals when d_{ii} does not exist as a fixed object. The agenda for Paper 5b and beyond.

The volume is designed to be read end-to-end by a working trade economist who already has Volume 1 internalized. Each chapter has a worked example, a notation parse on first appearance of a new symbol, and a caveat box where the assumptions are tighter than they look.

A note on style: I am the sole author of this volume. I write in the first person where my own analytical choices are at stake (“I take $\delta = 1$ because...”) and in plain declarative voice where the mathematics is doing the work. The volume is opinionated. When the literature has been wrong, the volume says so. The reader is welcome to disagree.

Part I

Numerical evaluation and regularization

Chapter 1

Why discretization matters for divergent integrals

1.1 The pole, restated

Volume 1, Chapter 11 established that the 2D moment integral $\int_0^R r^{\theta+1} dr$ diverges at the lower limit for $\theta \leq -2$. The reason is geometric: the pairwise-distance density of a uniform disk scales as $f(r) \sim r$ near the origin (this is the 2D version of the linear vanishing of the pdf at $r = 0$), so the integrand $r^\theta \cdot r$ behaves as $r^{\theta+1}$ near zero, and the integral converges if and only if $\theta + 1 > -1$, i.e., $\theta > -2$. Below the pole, the continuous integral has no finite value.

What does this mean operationally? It means that if you write down the Head-Mayer (2002) closed form

$$d_{ii}(\theta) = \left(\frac{2}{\theta + 2} \right)^{1/\theta} R$$

and try to evaluate it at $\theta = -3$, the formula gives a complex number on the principal branch. If you write down the integral itself and try to evaluate it numerically on a discretized region, you get a finite number, but the number depends on the discretization grain.

This chapter walks through what the discretized number is, how it depends on the grain, and what the resulting object actually represents.

1.2 The discretization

Suppose we have a uniform-density disk of radius R and we want to compute the CES expected pairwise distance at $\theta = -5$. The analytical integral diverges. Numerically, we proceed by sampling

N atoms uniformly within the disk and computing the empirical version:

$$\hat{d}_{ii}^{\text{eff}}(\theta = -5) = \left(\frac{1}{N(N-1)} \sum_{i \neq j} \|x_i - x_j\|^{-5} \right)^{-1/5}. \quad (1.1)$$

For N atoms in the disk, the smallest pairwise distance is roughly $R/\sqrt{N}$ (the average nearest-neighbor distance). Atoms closer than this are rare; the sum is dominated by the few near-collisions.

Notation parse

$d_{\min}$ is the regularization cutoff: the spatial grain below which we either don't sample atoms or we cap pairwise distances. Two natural implementations: (1) sample N atoms uniformly, accept whatever $d_{\min}$ falls out; (2) sample many atoms but explicitly cap $d_{ij} \mapsto \max(d_{ij}, d_{\min})$. Both give the same qualitative behavior; the second is easier to control.

1.3 What the discretized object measures

For $\theta > -2$ (above the pole), the empirical estimator $\hat{d}_{ii}^{\text{eff}}$ converges to the true continuous integral as $N \rightarrow \infty$ or $d_{\min} \rightarrow 0$. The discretization grain is a numerical-accuracy issue, not a conceptual one.

For $\theta \leq -2$, the empirical estimator does not converge. As N grows (or $d_{\min}$ shrinks), $\hat{d}_{ii}^{\text{eff}}$ decreases monotonically toward zero. There is no fixed point. The number you report at any specific $d_{\min}$ is a regularization choice tied to that grain, not a property of the underlying disk geometry.

The continuous CES expected pairwise distance at gravity-consistent θ is not a number. It is a function $d_{ii}^{\text{eff}}(\theta, d_{\min})$ defined for every $d_{\min} > 0$, with $d_{ii}^{\text{eff}} \rightarrow 0$ as $d_{\min} \rightarrow 0$. The CEPII evaluation at $\theta = -1$ exists because $\theta = -1$ is above the pole. Below the pole, CEPII would not give a single number either; it would give a sequence indexed by the city sample.

1.4 The empirical sweep

Notebook 19c (the regularization sensitivity script) runs the discretization at $\theta = -5$ on a uniform disk of radius $R = 1000$ km, varying $d_{\min} \in \{0.01, 0.1, 1, 5, 10, 50, 100, 200, 500\}$ km. The output:

$d_{\min}$ (km)	$\hat{d}_{ii}^{\text{eff}}$ (km)	ratio to HM at $\theta = -1$
0.01	0.063	0.0001
0.1	0.631	0.0013
1	6.288	0.013
5	29.4	0.059
10	51.9	0.10
50	149.7	0.30
100	229.4	0.46
200	352.9	0.71
500	640.4	1.28

The ratio to the HM-at- $\theta = -1$ value ($R/2 = 500$ km) crosses unity between $d_{\min} = 200$ km and $d_{\min} = 500$ km. At $d_{\min} = 500$ km, the discretized $\theta = -5$ result exceeds the $\theta = -1$ analytical result; at $d_{\min} = 0.01$ km, it is four orders of magnitude smaller. There is no fixed point. The implied d_{ii} is a function of $d_{\min}$, full stop.

1.5 What this implies for empirical practice

Three operational implications follow.

First, any specific value for d_{ii} reported in a gravity paper must be accompanied by a statement of the regularization grain. CEPII's $\theta = -1$ choice produces a single number for each country because $\theta = -1$ is above the pole. Below the pole, the single-number convention breaks down.

Second, papers that swap CEPII's distance for a satellite-calibrated alternative (the Gonchar-Helfrich panel, the WorldPop-recomputed distances, the OSM-based least-cost-path measures) are choosing one regularization over another, not solving a measurement problem. The Paper 5a critique applies to all of them.

Third, the structural-versus-reduced-form distinction in θ is not a notational hygiene matter. It is a substantive methodological choice with empirical consequences. Papers that report border-puzzle multipliers as structural objects, rather than as conditional on a regularization, have misrepresented the literature's empirical content.

Chapter 2

The regularization sweep on a uniform disk

2.1 Setup

This chapter reproduces the uniform-disk sweep in Python and walks through the implementation. The script is at `notebooks/19c_regularization_sensitivity.py` in the project repository.

The setup: place N atoms uniformly within a disk of radius $R = 1000$ km, compute all pairwise distances, cap pairwise distances at $d_{\min}$ from below, then evaluate the CES expected pairwise distance at $\theta = -5$. Repeat for 5 random seeds, average across seeds, and sweep $d_{\min}$.

The atoms are sampled by inverse-CDF: $r_i = R\sqrt{u_i}$ and $\phi_i \sim \text{Uniform}(0, 2\pi)$, with $u_i \sim \text{Uniform}(0, 1)$. The $\sqrt{}$ ensures the resulting density is uniform in 2D area; the naive sampling $r_i \sim \text{Uniform}(0, R)$ would over-sample near the center.

Cartesian coordinates: $x_i = r_i \cos \phi_i$, $y_i = r_i \sin \phi_i$. Pairwise distance: $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$.

The CES expected distance at $\theta = -5$:

$$\hat{d}_{ii}^{\text{eff}} = \left(\frac{1}{N(N-1)} \sum_{i \neq j} d_{ij}^{-5} \right)^{-1/5}.$$

The capped version: $\tilde{d}_{ij} = \max(d_{ij}, d_{\min})$, then the same formula.

2.2 Why the result is essentially linear in $d_{\min}$ at low $d_{\min}$

For θ well below the pole, the empirical sum is dominated by the pairs with the smallest distances. With N atoms and a uniform distribution, the expected number of pairs with $d < d_{\min}$ scales as $N^2 \cdot (d_{\min}/R)^2$. The sum $\sum d_{ij}^{-5}$ over these pairs scales as $N^2 \cdot (d_{\min}/R)^2 \cdot d_{\min}^{-5} = N^2 \cdot d_{\min}^{-3} \cdot R^{-2}$. Dividing by $N(N-1) \approx N^2$ and taking the $(-1/5)$ -th root:

$$\hat{d}_{ii}^{\text{eff}} \sim (d_{\min}^{-3} R^{-2})^{-1/5} = d_{\min}^{3/5} R^{2/5}.$$

So $\hat{d}_{ii}^{\text{eff}}$ scales as $d_{\min}^{3/5}$ at low $d_{\min}$: sub-linear but with a sharp slope. The simulation output (Section 1.4) fits this scaling.

The same scaling argument generalizes: for $\theta < -2$, the scaling is $\hat{d}_{ii}^{\text{eff}} \sim d_{\min}^{(\theta+2)/(\theta)}$ at small $d_{\min}$. At $\theta = -3$: exponent $1/3$. At $\theta = -5$: exponent $3/5$. At $\theta = -7$: exponent $5/7$. The implied d_{ii} flattens out (approaches a finite limit only weakly) as θ becomes more negative.

The slope of $\log \hat{d}_{ii}^{\text{eff}}$ versus $\log d_{\min}$ at small $d_{\min}$ identifies the scaling exponent $\alpha(\theta) = (\theta+2)/\theta$. At $\theta = -1$ (above the pole), $\alpha = -1$: the implied distance INCREASES as $d_{\min}$ decreases. At $\theta = -2$ (the pole), $\alpha = 0$: log-flat. At $\theta = -3$, $\alpha = 1/3$: weak dependence. At $\theta = -7$, $\alpha = 5/7$: near-linear dependence. The pole at $\theta = -2$ is the sign-change of the regularization-elasticity.

2.3 What real country polygons add

The uniform disk is a clean theoretical object. Real countries are not uniform disks. They have irregular shape (e.g., Norway, Indonesia, Chile), internal water bodies (e.g., the US Great Lakes), and non-uniform population density (every country, but extreme cases like Russia and Canada). The regularization curve on a real country polygon will generally not match the uniform-disk curve.

Chapter 14 returns to this. For now, the uniform-disk result establishes the scaling regime; real-country corrections are quantitative deviations from it, not qualitative departures.

Chapter 3

Real-country polygons and the Mollweide projection

3.1 Why projections matter for area

[STATUS: skeleton — full development in v0.2 of Volume 2.]

Pages of text on equal-area projections, the Mollweide ESRI:54009 projection used in the Paper 5a pipeline, comparison with Web Mercator (common but bad for area), and the resulting area errors for high-latitude countries. The full chapter would walk through the `geopandas.to_crs("ESRI:54009")` pipeline.

3.2 Sampling atoms within a real polygon

[STATUS: skeleton.]

Rejection sampling: draw points in the polygon's bounding box, reject those outside the polygon, accept those inside. With N accepted atoms, the rejection rate equals (bounding box area) / (polygon area). For a country with low fill factor (e.g., Indonesia with all its archipelagos), rejection sampling is inefficient. Triangulation-based sampling (triangulate the polygon, sample uniformly within each triangle weighted by triangle area) is more efficient.

3.3 The country-by-country regularization sweep

[STATUS: skeleton.]

Full development would tabulate $d_{ii}^{\text{eff}}(\theta = -5, d_{\min})$ for the 27-country pilot subset (USA, RUS, CAN, BRA, AUS, CHN, IND, JPN, FRA, DEU, NLD, BEL, KOR, GBR, ITA, ESP, MEX, ARG, CHL, EGY, plus 7 smaller countries). The cross-country variation in the regularization curve tells us how much the

small-large country geometry affects the mechanism.

Part II

Real-data construction and the GIS pipeline

Chapter 4

BACI and the international trade panel

4.1 What BACI is and why we use it

BACI is CEPII's reconciled bilateral trade panel, derived from UN Comtrade with consistency adjustments. The raw Comtrade data has the familiar reporting problem: country i records its exports to country j , country j records its imports from country i , and the two numbers disagree (sometimes by factors of 2). BACI applies a reconciliation algorithm that produces a single best-estimate bilateral flow X_{ij} for each ordered pair, by year and HS6 product code.

The current versions:

- **BACI HS02 v202401b**: HS 2002 nomenclature, vintage of January 2024, “b” suffix indicating the second revision of that vintage. Covers 1995-2022 (most years). What Paper 5a uses.
- **BACI HS17 v202401**: HS 2017 nomenclature, newer products (e.g., LNG, electric vehicles, drone components), shorter time series (2018 onward).
- **BACI HS92, HS96, HS07, HS12**: legacy vintages.

For a paper that needs a long time series and isn't focused on recent high-tech goods, HS02 is the right choice. The trade-off is that HS02's classifications are stale for goods that emerged after 2002.

4.2 File structure on disk

Each year is a separate CSV. The columns (per CEPII's data manual):

- **t**: year (integer)
- **i**: exporter ISO numeric code (3-digit integer; e.g., 840 = USA)
- **j**: importer ISO numeric code
- **k**: HS6 product code (6-digit integer)

- v: trade value in current thousand USD
- q: quantity (often missing; reported in metric tons or units)

A single year's file (e.g., BACI_HS02_Y2010_V202401b.csv) is around 200-300 MB and contains 5-10 million rows. The file is country-pair-product-year level. For a country-pair-year gravity panel, you aggregate over the product dimension.

Building the 2010 country-pair panel from BACI:

```

1 import pandas as pd
2 from pathlib import Path
3
4 f = Path("BACI_HS02_Y2010_V202401b.csv")
5
6 # Chunked read: aggregate by (exporter, importer) on the fly
7 pieces = []
8 for chunk in pd.read_csv(f, usecols=["i", "j", "v"],
9                             chunksize=1_000_000):
10     # Drop missing trade-value rows
11     chunk["v"] = pd.to_numeric(chunk["v"], errors="coerce")
12     chunk = chunk.dropna(subset=["v"])
13     # Aggregate within chunk
14     piece = chunk.groupby(["i", "j"], as_index=False)["v"].sum()
15     pieces.append(piece)
16
17 # Combine chunks and re-aggregate
18 agg = pd.concat(pieces).groupby(["i", "j"], as_index=False)["v"].sum()
19
20 # Convert to USD (BACI is in 1000 USD)
21 agg["X_ij_usd"] = agg["v"] * 1000
22
23 print(f"{len(agg):,} country-pair flows in 2010")
24 print(f"Total exports: ${agg['X_ij_usd'].sum() / 1e12:.1f}T")

```

For 2010, this gives about 30,605 country-pair flows totaling ~ \$15T in world trade.

4.3 Country-code mapping

BACI uses ISO 3166-1 numeric codes (840 for USA, 156 for China, etc.). CEPII Gravity uses ISO 3166-1 alpha-3 codes (USA, CHN, etc.). The mapping is a CEPII-provided lookup table; in practice, you can build it from the CEPII Gravity file itself:

```

1 cepii = pd.read_csv("Gravity_V202211.csv") # via chunked read for size
2 mapping = cepii[["iso3num_o", "iso3_o"]].dropna().drop_duplicates(
3     subset=["iso3num_o"])

```

```

4 num_to_iso = {int(n): s for n, s in mapping.values}
5
6 # Apply to BACI
7 agg["iso3_o"] = agg["i"].map(num_to_iso)
8 agg["iso3_d"] = agg["j"].map(num_to_iso)

```

A small number of BACI rows will have ISO numeric codes that don't match any CEPII alpha-3 (e.g., disputed territories, historical codes). Drop these or document them; they amount to less than 1% of trade value.

4.4 Zero-trade rows

A key issue for the OLS-versus-PPML choice: the BACI country-pair panel does NOT include rows where $X_{ij} = 0$. BACI only records positive trade. To build a panel with zero-trade rows for PPML, you merge BACI onto the full Cartesian product (CEPII Gravity is already the full Cartesian product for each year, including zero-trade pairs):

```

1 # Full CEPII panel for the year
2 inter = cepii[cepii["iso3_o"] != cepii["iso3_d"]][
3     ["iso3_o", "iso3_d", "gdp_o", "gdp_d", "distw_harmonic", "contig"]
4 ]
5
6 # Left-join BACI flows; missing => zero trade
7 inter = inter.merge(agg[["iso3_o", "iso3_d", "X_ij_usd"]],
8                     on=["iso3_o", "iso3_d"], how="left")
9 inter["X_ij_usd"] = inter["X_ij_usd"].fillna(0.0)
10
11 print(f"Total rows: {len(inter):,}")
12 print(f"Zero-trade rows: {(inter['X_ij_usd'] == 0).sum():,} ({100 * (inter['X_ij_usd'] == 0).mean():.1f}%)")

```

For 2010, this gives 63,234 inter-national rows of which about 32,000 (50%) are zero-trade. The presence of so many zeros is why PPML is the recommended estimator in the modern gravity literature.

The Wei (1996) intra-national flow proxy $X_{ii} = \max(0, \text{GDP}_i - \sum_j X_{ij})$ will be zero or close to zero for countries that export more than their GDP (a sign of re-export or merchanting, e.g., Singapore, Hong Kong, the Netherlands). These countries get dropped from the intra-national panel because their Wei proxy is zero. This is a substantive limitation that the Yotov-Piermartini-Monteiro-Larch (2016) “trade for production” proxy partially addresses by computing X_{ii} from sectoral production rather than total GDP.

4.5 Aggregating across HS6 products

For a country-pair-year gravity regression, you typically aggregate across HS6 products. For a sectoral gravity regression (e.g., Caliendo-Parro style), you keep the HS6 dimension and run separate regressions per sector. BACI is the right source for both.

A nuance: BACI's reconciliation is done at the HS6 level. If you aggregate to a coarser level (HS2 or HS-section) BEFORE running BACI's reconciliation, you get different numbers than aggregating AFTER. The canonical workflow is: keep BACI's HS6-level numbers, aggregate AFTERWARD if you don't need the sectoral dimension.

Chapter 5

CEPII Gravity v202211 and the Wei (1996) proxy

5.1 What CEPII Gravity contains

CEPII Gravity is the standard pair-year covariate file for structural gravity estimation. The current version, v202211 (released November 2022), contains 91 variables over 250 countries from 1948-2020, in a single Cartesian-product panel of 13.7 million rows (250 origins $\times$ 250 destinations $\times$ 60-70 years, with some pair-year gaps).

The variables relevant to Paper 5a:

- `iso3_o`, `iso3_d`: country alpha-3 codes
- `iso3num_o`, `iso3num_d`: country numeric codes (for joining to BACI)
- `year`: integer
- `gdp_o`, `gdp_d`: GDP in current USD (origin and destination)
- `pop_o`, `pop_d`: population
- `dist`: simple distance between capital cities, km
- `distcap`: distance between capital cities (synonym in some vintages)
- `distw_arithmetic`: arithmetic-mean of city-pairwise distances weighted by city population, km. This is the Mayer-Zignago (2011) “distwarith” variable.
- `distw_harmonic`: harmonic-mean of the same city-pairwise distances. This is the Mayer-Zignago “distwces” variable, evaluated at $\theta = -1$. **This is the variable Paper 5a critiques.**
- `contig`: 1 if countries share a land border, 0 otherwise
- `comlang_off`: 1 if countries share an official language
- `comlang_ethno`: 1 if at least 9% of the population speaks the same language in both countries

- `col_dep_ever`: 1 if there was a colonial relationship historically
- `rta`: 1 if there is a regional trade agreement in force in that year

For the intra-national rows (`iso3_o == iso3_d`), CEPII fills `distw_arithmetic` with the within-country arithmetic-mean of pairwise city distances and `distw_harmonic` with the within-country harmonic-mean. These are the Head-Mayer (2002) closed form evaluated at $\theta = +1$ and $\theta = -1$ respectively.

5.2 Reading the CSV

The 13.7M-row file is approximately 4 GB in CSV form. On a FAT32 file system (e.g., an external USB drive), the file is at or near the 4 GB per-file limit, which can cause read errors. Best practice: store the file on a local SSD or, even better, convert to Parquet:

```
1 import pandas as pd
2 import pyarrow.parquet as pq
3
4 # One-time conversion
5 df = pd.read_csv("Gravity_V202211.csv")
6 df.to_parquet("Gravity_V202211.parquet", compression="snappy")
```

Parquet reads are 5-10x faster and the file is 700 MB (compressed). For a year-filtered read (e.g., 2010 only), chunked CSV reads are also acceptable:

```
1 def chunked_cepii(year):
2     chunks = []
3     for chunk in pd.read_csv("Gravity_V202211.csv",
4                             dtype={"iso3_o": str, "iso3_d": str, "year": "Int64"},
5                             chunksize=500_000, low_memory=False):
6         sub = chunk[chunk["year"] == year]
7         if len(sub) > 0:
8             chunks.append(sub)
9     return pd.concat(chunks, ignore_index=True)
```

For 2010, a chunked read produces a 63,504-row DataFrame (17 MB) in about 3-5 minutes on a typical laptop with the CSV on an external drive.

5.3 The Wei (1996) intra-national flow proxy

For intra-national trade (no analog of bilateral exports in trade statistics), Wei (1996) proposed the residual proxy

$$X_{ii,t} = \max(0, Y_{i,t} - \sum_{j \neq i} X_{ij,t}) \quad (5.1)$$

where $Y_{i,t}$ is country i 's GDP in year t and the sum is over its exports to all other countries. The intuition: total GDP equals total production, which is sold either domestically or as exports, so the residual is intra-national sales.

The Wei proxy is the standard in the literature but has known limitations.

Wei proxy limitations:

1. **Re-exports and merchanting** (Singapore, Hong Kong, Netherlands, Belgium). When a country's exports exceed its domestic production (because it re-exports goods produced elsewhere), the Wei proxy gives zero or negative. The $\max(0, \cdot)$ truncation produces a zero, which excludes the country from the intra-national panel.
2. **Sectoral composition.** GDP includes services (70% of US GDP) that are largely non-tradable. The Wei proxy treats services as intra-national trade, but they're not really "traded" in the gravity-model sense.
3. **Trade in second-hand goods, intra-firm transfers, and missing-flow estimation.** Comtrade doesn't capture all flows.

The Yotov-Piermartini-Monteiro-Larch (2016) "trade for production" proxy uses sectoral GDP (specifically, gross output minus exports within each tradable sector), which addresses the services and re-export issues. It requires sectoral data that's harder to obtain but is the modern standard.

5.4 Building the intra-national rows from CEPII + BACI

```

1 # Intra-national rows construction
2 gdp_lookup = dict(
3     cepii[["iso3_o", "gdp_o"]]
4     .drop_duplicates().dropna().values
5 )
6
7 exports_by_origin = (
8     baci.groupby("iso3_o", as_index=False)["X_ij_usd"].sum()
9     .set_index("iso3_o")["X_ij_usd"].to_dict()
10 )
11
12 # Pull CEPII's stored intra-national distance
13 cepii_self = cepii[cepii["iso3_o"] == cepii["iso3_d"]][
14     ["iso3_o", "distw_harmonic", "distw_arithmetic"]
15 ].copy()
16

```

```

17 intra_rows = []
18 for iso, gdp in gdp_lookup.items():
19     if not pd.notna(gdp) or gdp <= 0:
20         continue
21     exp = exports_by_origin.get(iso, 0.0)
22     x_ii = max(0.0, gdp - exp)
23     intra_rows.append({
24         "iso3_o": iso, "iso3_d": iso,
25         "X_ij_usd": x_ii,
26         "gdp_o": gdp, "gdp_d": gdp,
27         "home": 1,
28     })
29
30 intra_df = pd.DataFrame(intra_rows)
31 intra_df = intra_df.merge(cepii_self, on="iso3_o", how="left")
32 print(f"Intra rows: {len(intra_df):,}")
33 print(f"With positive Wei proxy: {(intra_df['X_ij_usd'] > 0).sum():,}")
34 print(f"With CEPII intra distance: {intra_df['distw_harmonic'].notna().sum():,}")

```

For 2010, this gives 218 candidate intra rows, of which 10 have BOTH positive Wei proxy AND non-missing CEPII intra distance. The other 208 are dropped (re-export countries, missing data). The small $n_{\text{intra}} = 10$ is the source of Paper 5a’s identification caveat in Section 5.3.

5.5 Why the 10 surviving countries are large economies

The 10 intra-national rows for 2010 typically include: USA, China, Japan, Germany, France, UK, Italy, Brazil, Canada, Russia. These are all large economies with GDP well above their export totals. Smaller countries like the Netherlands (large exports relative to GDP from re-exports) or open-economy specialists like Singapore drop out.

This selection effect is itself worth thinking about. The “intra-national” panel that the border-puzzle literature has been estimating against is dominated by a few G7 economies. Generalizations to “borders matter for trade” are really “borders matter for the particular set of large economies that survive the Wei filter.”

5.6 Joining BACI and CEPII into a panel

The final gravity panel is the union of the inter-national rows (from CEPII, with X_{ij} filled in from BACI) and the intra-national rows (constructed from the Wei proxy). For OLS, you keep only positive X_{ij} ; for PPML, you keep zero-trade rows.

```

1 # Inter-national rows
2 inter = cepii[cepii["iso3_o"] != cepii["iso3_d"]][
3     ["iso3_o", "iso3_d", "gdp_o", "gdp_d",

```



```

4     "distw_harmonic", "contig", "comlang_off"]
5 ].copy()
6 inter = inter.merge(
7     baci[["iso3_o", "iso3_d", "X_ij_usd"]],
8     on=["iso3_o", "iso3_d"], how="left"
9 )
10 inter["X_ij_usd"] = inter["X_ij_usd"].fillna(0.0) # zeros for PPML
11 inter["home"] = 0
12
13 # Combine intra + inter
14 panel = pd.concat([inter, intra_df], ignore_index=True, sort=False)
15 print(f"Total panel: {len(panel):,} rows")
16 print(f"Intra-national: {int(panel['home'].sum()):,}")
17 print(f"Positive flows: {(panel['X_ij_usd'] > 0).sum():,}")
18 print(f"Zero flows: {(panel['X_ij_usd'] == 0).sum():,}")

```

For 2010: 63,452 total rows, 10 intra-national, 28,053 positive flows (the OLS sample), 35,399 zero flows (PPML uses these too).

Chapter 6

Population and activity rasters: GHS-POP, WorldPop, VIIRS

6.1 Why rasters matter for intra-national distance

The Mayer-Zignago (2011) CEPII intra-national distance is a discretization of the underlying CES integral, with the discretization implemented as a population-weighted harmonic mean over city centroids. Their city sample comes from a circa-2004 cut of the World Gazetteer city list, with ~ 25 city atoms per country on average and large coverage gaps. The natural alternative, given modern data, is to replace the city subsample with a population raster: a regular grid of cells where each cell carries a population (or activity) weight. This is the construction at the heart of the Gonchar-Helfrich (2026) effective-distance panel.

Three rasters are relevant for Paper 5a's empirical strategy.

6.2 GHS-POP (JRC Global Human Settlement)

The **Global Human Settlement Population grid** is produced by the European Commission Joint Research Centre. It comes in two spatial resolutions:

- GHS-POP at 100 m: the canonical fine-resolution global population raster. Each $100\text{ m} \times 100\text{ m}$ cell carries a population count.
- GHS-POP at 1 km: aggregated version, easier to handle, sufficient for country-aggregate analysis.

Available years: 1975, 1980, 1985, 1990, 1995, 2000, 2005, 2010, 2015, 2020. The 5-year cadence is fine for cross-section but limiting for panel work; for year-by-year coverage, WorldPop is the alternative.

The raster is delivered as a GeoTIFF in the Mollweide equal-area projection (ESRI:54009). The Moll-

weide choice is important: at high latitudes, naive lat-lon rasters distort area significantly (Greenland looks bigger than Africa in Mercator). Mollweide preserves area exactly, which is what you need when computing population-weighted spatial statistics.

Reading GHS-POP and masking by country:

```

1 import rasterio
2 from rasterio.mask import mask
3 from rasterstats import zonal_stats
4 import geopandas as gpd
5
6 # Country boundaries (GADM, in Mollweide for consistency)
7 bnd = gpd.read_file("gadm_410_level0.gpkg").to_crs("ESRI:54009")
8 us = bnd[bnd["GID_0"] == "USA"]
9
10 # Mask GHS-POP raster by US polygon
11 with rasterio.open("GHS_POP_E2020_GLOBE_R2023A_54009_1000_V1_0.tif") as src:
12     img, transform = mask(src, us.geometry, crop=True, filled=False)
13     nodata = src.nodata or 0
14     img = img.filled(nodata)
15
16 # Aggregate population
17 us_pop = img[img > 0].sum()
18 print(f"US 2020 population: {us_pop / 1e6:.1f}M") # ~331M

```

6.3 WorldPop (annual updates 2000-2024)

WorldPop is a Southampton-led project providing annual population rasters at 100 m and 1 km resolution from 2000 onward. The 100 m version uses a Bayesian dasymetric mapping approach to redistribute census counts based on building footprints, road density, and nighttime lights. It is essentially the highest-resolution global population raster available at annual cadence.

For a Paper 5a-style regularization sweep, WorldPop's annual cadence is overkill (one year per regression is sufficient). For Gonchar-Helfrich's 2000-2024 panel, the annual cadence is essential.

6.4 VIIRS Day/Night Band (nighttime lights)

The NOAA VIIRS Day/Night Band on the Suomi-NPP satellite (and later NOAA-20 and NOAA-21) produces nightly composites of nighttime light radiance at ~ 750 m resolution from 2012 onward. The annual composites are at eogdata.mines.edu.

Why use VIIRS? Nighttime lights are a well-validated proxy for economic activity (Henderson,

Storeygard, Weil 2012, AER). Where population rasters tell you where people live, VIIRS tells you where they're producing, consuming, and lighting up at night. For a "population-weighted" versus "activity-weighted" comparison of intra-national distance, you need both.

6.5 Working with rasters in Python

The standard toolchain:

- `rasterio` for reading/writing GeoTIFFs
- `rasterstats.zonal_stats` for aggregating raster values within polygons
- `geopandas` for vector data (country boundaries, polygons)
- `xarray` + `rioxarray` for label-aware raster manipulation

Projection consistency matters. If your boundaries are in EPSG:4326 (lat-lon) and your raster is in ESRI:54009 (Mollweide), `rasterio.mask.mask` will refuse the operation (or, worse, silently misalign). Always check `src.crs` on the raster and `gdf.crs` on the vector, and reproject one to match the other.

6.6 What changes when you use a raster instead of CEPII's cities

CEPII's city-based intra-national distance, at $\theta = -1$, gives USA $\approx 1,177$ km. The raster-based equivalent, at $\theta = -1$, on GHS-POP 2020 at 1 km resolution, gives USA $\approx 1,140$ km (a 3% difference). The two methodologies are operationally similar at $\theta = -1$, *above the pole*.

At $\theta = -5$, the two methodologies diverge sharply. The CEPII city sample produces a number (whatever the harmonic-mean at $\theta = -5$ happens to be on the city subsample). The raster sample produces a number that depends on the raster resolution. There is no agreement, because below the pole there is no fixed truth.

This is the central empirical insight of Paper 5a: the choice of underlying spatial data (CEPII cities vs. GHS-POP raster vs. WorldPop raster vs. a hypothetical even-finer-grain population layer) is the regularization choice $d_{\min}$. CEPII's cities have effective $d_{\min} \approx 50 - 100$ km (city centers are typically separated by that distance); GHS-POP at 1 km has $d_{\min} = 1$ km; WorldPop at 100 m has $d_{\min} = 0.1$ km. The implied d_{ii}^{eff} falls across these choices because the integral has no fixed point.

Chapter 7

Transport-network distance: OpenStreetMap, ports, and air routes

[STATUS: skeleton.] Coverage of how to construct a multi-modal transport graph from OpenStreetMap road data, World Port Index (WPI) maritime data, and OpenFlights air routes. The Dijkstra least-cost-path solver in `networkx` or (faster) Rust via `paper5-core`. The Gonchar-Helfrich panel's transport-distance variant.

Part III

Regression specification

Chapter 8

OLS gravity with origin and destination fixed effects

8.1 The canonical specification

The workhorse log-OLS gravity regression for cross-section trade data is:

$$\log X_{ij,t} = \alpha + \beta \log d_{ij,t} + \gamma \cdot \mathbf{1}\{i = j\} + \eta_{i,t} + \mu_{j,t} + \epsilon_{ij,t}. \quad (8.1)$$

The variables:

- $X_{ij,t}$: bilateral trade from origin i to destination j in year t .
- $d_{ij,t}$: bilateral distance (CEPII's `distw_harmonic` typically).
- $\mathbf{1}\{i = j\}$: the home dummy. 1 for intra-national rows (Wei proxy), 0 for inter-national.
- $\eta_{i,t}, \mu_{j,t}$: origin-year and destination-year FE.
- $\epsilon_{ij,t}$: error term, assumed mean-zero conditional on regressors.

The coefficient of substantive interest is γ , the home-bias multiplier in log units. $e^{\hat{\gamma}}$ is the ratio of intra-national to equivalent-distance international trade, conditional on the FE and distance.

8.2 What the FE absorb

In a single-year cross-section, the origin FE η_i absorbs:

- Origin's GDP (constant per origin in a given year).
- Origin's population (constant per origin in a given year).
- Origin's average tariff level, average exchange rate, currency composition: anything that doesn't vary across destinations within the origin.

- In a structural reading, the origin's outward multilateral resistance term Π_i .

Similarly μ_j absorbs destination-side characteristics including the inward multilateral resistance P_j . The Anderson-van Wincoop (2003) insight is that these multilateral resistance terms are why the McCallum (1995) raw estimate of $22\times$ inflates the structural border effect; properly absorbed, the multiplier shrinks to $\sim 5\times$.

A common student mistake is to include $\log Y_i$ (origin GDP) or $\log Y_j$ (destination GDP) as explicit regressors alongside the country FE. In a single-year cross-section, these are perfectly collinear with the FE structure and should be omitted (or you should omit the FE and include the GDP terms). Including both is the collinearity trap of Chapter 9.

8.3 What the home dummy identifies

For an intra row ($\text{iso_o} = \text{iso_d} = \text{USA}$, say):

$$\log \hat{X}_{\text{USA}, \text{USA}} = \alpha + \beta \log d_{\text{USA}, \text{USA}} + \gamma + \hat{\eta}_{\text{USA}} + \hat{\mu}_{\text{USA}}.$$

For an inter row ($\text{iso_o} = \text{USA}$, $\text{iso_d} = \text{CAN}$):

$$\log \hat{X}_{\text{USA}, \text{CAN}} = \alpha + \beta \log d_{\text{USA}, \text{CAN}} + \hat{\eta}_{\text{USA}} + \hat{\mu}_{\text{CAN}}.$$

The home dummy γ is identified from the residual difference between intra-row predictions and inter-row predictions, after absorbing the origin and destination FE. Specifically, $\hat{\gamma}$ captures the average log-difference between intra-row observed flows and what the inter-row fit predicts for the intra-row distance.

8.4 Why $\hat{\gamma}$ has high variance with small n_{intra}

In a typical 2010 BACI-CEPII panel, you have $\sim 28,000$ inter-national observations and ~ 10 intra-national observations. The FE structure absorbs ~ 500 degrees of freedom (250 origins + 250 destinations, minus 1 for the constant). The home dummy is identified from the intra rows after the FE absorb the country-specific levels.

With 10 intra observations and 500 FE parameters that can shift to absorb intra-row residuals, the home dummy has thin identification. The typical standard error is ~ 0.7 in log units, which is substantial. For Paper 5a, the within-year movement of $\hat{\gamma}$ across $d_{\min}$ is large (~ 3 log units) relative to the standard error, so the regularization-sensitivity finding is robust. But the across-year level comparison (year 2010 vs. 2015) is not directly interpretable, because the set of countries that survive the Wei filter differs across years.

Estimating the canonical gravity regression in Python:

```

1 import pandas as pd
2 import statsmodels.formula.api as smf
3 import numpy as np
4
5 # panel: pd.DataFrame with columns iso3_o, iso3_d, distance, X_ij_usd, home
6 df = panel.copy()
7 df = df[(df["X_ij_usd"] > 0) & (df["distance"] > 0)]
8 df["log_X"] = np.log(df["X_ij_usd"])
9 df["log_d"] = np.log(df["distance"])
10
11 # Formula API (Patsy handles dummy construction)
12 formula = "log_X ~ log_d + home + C(iso3_o) + C(iso3_d)"
13 model = smf.ols(formula, data=df).fit()
14
15 print(f"log_d coefficient: {model.params['log_d']:+.4f}")
16 print(f"home coefficient: {model.params['home']:+.4f}")
17 print(f"home multiplier: {np.exp(model.params['home']):.2f}x")
18 print(f"R^2: {model.rsquared:.3f}")

```

For year 2010 with CEPII's intra-distance, this gives $\hat{\beta}_{\log d} = -1.836$, $\hat{\gamma} = +2.132$, multiplier $8.43\times$. The McCallum-AvW range.

8.5 The case for and against OLS

OLS works when:

- Zero-trade rows are absent (or you're willing to drop them).
- The conditional mean is well-described by a log-linear specification.
- You're estimating $\mathbb{E}[\log X|Z]$ rather than $\log \mathbb{E}[X|Z]$.

OLS fails when:

- Zero-trade rows are a substantial fraction (50% in our 2010 BACI panel). Dropping them induces a selection bias.
- Heteroskedasticity is severe and depends on covariates. The Silva-Tenreyro (2006) point: $(X_{ij}|Z)$ scales with $E[X_{ij}|Z]^2$ in many gravity settings, which gives biased OLS.

PPML (Chapter 10) addresses both failure modes. For Paper 5a's structural critique, the OLS specification is documented as the primary; PPML is a robustness check.

Chapter 9

The perfect-collinearity trap: log-GDP regressors and country FE

This chapter documents a methodological pitfall I encountered while running the Paper 5a empirical sweep. It is included in the volume because the lesson generalizes beyond this specific paper.

9.1 The bug

I built a Spec B-prime regression with the design matrix constructed manually via `statsmodels.api.OLS`, including `log_gdp_o`, `log_gdp_d`, `contig`, `comlang_off` as columns alongside origin and destination one-hot dummies. The baseline home coefficient came out as -4.96 for year 2010 (multiplier 0.007, no border puzzle at all), inconsistent with a known-working version that produced $+2.32$ (multiplier 10, in the McCallum-AvW range).

9.2 The diagnosis

`statsmodels.api.OLS` does NOT automatically drop perfectly collinear columns. When confronted with a rank-deficient design matrix, it falls back to the Moore-Penrose pseudo-inverse. The pseudo-inverse gives a coefficient vector that minimizes the L2-norm subject to the fit being least-squares optimal, but the coefficient assignment among collinear columns is arbitrary.

In a single-year cross-section, `log_gdp_o` is perfectly collinear with the origin one-hot dummy structure (one GDP value per origin), and `log_gdp_d` with the destination dummies. The pseudo-inverse redistributes the colinear contribution across the collinear set, including the home dummy because of numerical noise in the singular-value decomposition.

9.3 The fix

`statsmodels.formula.api.ols` uses the formula API (R-style syntax), which builds the design matrix via Patsy. Patsy detects perfect collinearity at the dummy-construction stage and drops the redundant columns. The same regression with the formula API:

```
1 import statsmodels.formula.api as smf
2 formula = "log_X ~ log_d + home + C(iso3_o) + C(iso3_d)"
3 model = smf.ols(formula, data=df).fit()
```

gives the correct +2.13 home coefficient.

In single-year cross-sectional regressions with country FE, do not include log-GDP or other origin-constant / destination-constant regressors alongside the FE. The country FE absorb all single-side variation; the log-GDP terms add no information and corrupt the non-collinear coefficients via the pseudo-inverse fallback. Use the formula API or pre-drop the collinear columns. This is not a `statsmodels` bug; it is a design choice that requires the user to handle collinearity explicitly when using `sm.OLS`.

9.4 Generalizations

The same trap applies in any cross-sectional regression with high-dim fixed effects and additional regressors that are perfectly collinear with the FE structure. Examples:

- Country FE plus country-level log-GDP, log-population, capital-output ratio.
- Industry FE plus industry-level Herfindahl, average TFP, capital intensity.
- Year FE plus year-level interest rate, inflation, recession dummy (these vary by year only).

The general rule: if a regressor takes one value per FE category, do not include both. The FE absorbs the regressor.

In panel regressions with multiple years, log-GDP varies within FE categories (each country has multiple GDP observations across years). The collinearity is broken. The pitfall is specific to single-year cross-sections.

Chapter 10

Poisson PML and the Silva-Tenreyro (2006) zero-trade problem

10.1 What Silva-Tenreyro (2006) showed

The Silva-Tenreyro (2006) ReStat paper is the most-cited methodological paper in modern gravity estimation. The result, in plain English: log-OLS gravity is biased when:

1. There are zero-trade rows that get dropped from the log transformation.
2. The conditional variance ($X|Z$) scales with the conditional mean $\mathbb{E}[X|Z]$ (typical for trade data).

The bias is upward in absolute value of the distance elasticity: log-OLS tends to overstate the elasticity of trade with respect to distance. With zero rows present, the bias can be 30-50%.

The Silva-Tenreyro fix: estimate

$$\mathbb{E}[X_{ij,t}|Z_{ij,t}] = \exp(\alpha + \beta \log d_{ij,t} + \gamma \mathbf{1}\{i = j\} + \eta_{i,t} + \mu_{j,t}) \quad (10.1)$$

by Poisson Pseudo-Maximum Likelihood (PPML). The estimator is consistent under the conditional-mean assumption alone, not requiring $X_{ij,t}$ to actually follow a Poisson distribution. The “pseudo-maximum-likelihood” qualifier captures this: we use the Poisson likelihood as a quasi-likelihood criterion.

10.2 What PPML estimates

By Jensen’s inequality, $\log \mathbb{E}[X|Z] \neq \mathbb{E}[\log X|Z]$ whenever $(X|Z) > 0$. Log-OLS estimates the right-hand-side conditional mean; PPML estimates the left-hand-side. These are different objects. They have different coefficients.

For the gravity context: $\hat{\beta}_{\log d}$ from log-OLS is typically -1.5 to -2.0 ; $\hat{\beta}_{\log d}$ from PPML is typically -0.8 to -1.0 . The interpretation: trade flows decline with distance, but the rate of decline depends on what you're estimating.

10.3 The home coefficient under PPML versus OLS

This is the part that surprised me in Paper 5a's empirical work, so I'll be explicit. For the same 2010 BACI-CEPII panel:

- OLS gives $\hat{\gamma} = +2.13$, multiplier $8.4\times$ (in the McCallum-AvW range).
- PPML (preliminary, with convergence warnings) gives $\hat{\gamma} = -5.01$, multiplier $0.007\times$.

These are not contradictory; they are estimates of different objects. PPML estimates the conditional mean of intra-trade given the gravity covariates; OLS estimates the conditional mean of log-intra-trade. The Jensen-inequality wedge can be large for high-variance distributions like trade flows, and the wedge depends on the variance structure.

What survives across both: the regularization-sensitivity pattern. As $d_{\min}$ shrinks, $\hat{\gamma}$ becomes more negative under both estimators. The Paper 5a finding "the home coefficient is regularization-driven" is robust to estimator choice; the precise magnitude is not.

10.4 PPML in statsmodels

The Python implementation:

```
1 import statsmodels.formula.api as smf
2 import statsmodels.api as sm
3
4 formula = "X_ij_usd ~ log_d + home + C(iso3_o) + C(iso3_d)"
5 model = smf.glm(
6     formula, data=df,
7     family=sm.families.Poisson(link=sm.families.links.Log())
8 ).fit(method="bfgs", maxiter=200, tol=1e-6)
```

statsmodels.GLM is NOT the right tool for high-dimensional FE PPML. With ~ 500 FE parameters (one-hot encoded), the BFGS optimizer struggles to converge. Convergence warnings appear on every cell of a regularization sweep, and the coefficient estimates are the optimizer's last iterate rather than the true MLE.

The canonical implementation is the Correia-Guimaraes-Zylkin (2020) `ppmlhdfc` Stata package, which uses an absorbing-FE estimator that re-parameterizes the model in a way that avoids the

high-dimensional design matrix. The Python equivalent is `pyhdfe` + a custom IRLS loop, or `fixest` via `rpy2`. For Paper 5a, I report `statsmodels.GLM` results as preliminary and flag the need for `ppmlhdfe` as the natural follow-up.

10.5 Why this matters for Paper 5a

Paper 5a’s structural claim does not depend on the estimator. The integration parameter $\theta = \delta(1 - \sigma)$ is structural; the divergence of the integral at $\theta \leq -2$ is mathematical; the regularization-dependence of any specific number assigned to d_{ii} is a feature of the problem, not of the estimator.

Paper 5a’s empirical claim — the home coefficient depends on the regularization grain — is also robust to estimator choice. Both OLS and PPML produce a monotonic relationship between $\hat{\gamma}$ and $d_{\min}$, with sign and direction matching.

What changes between estimators is the level of $\hat{\gamma}$ at the CEPII baseline. OLS gives +2.1 (in the McCallum range); PPML gives −5 (a sign flip). Both are legitimate estimates of different objects. The paper reports OLS as primary because it is interpretable in the McCallum-AvW border-puzzle tradition; PPML is reported as a robustness check with the appropriate caveats.

Chapter 11

Pair-year fixed effects: the YPMM cookbook

[STATUS: skeleton.] The Yotov-Piermartini-Monteiro-Larch (2016) advanced guide to trade-policy analysis. The three-way fixed-effect specification: exporter-year FE, importer-year FE, pair FE. Why pair FE absorbs the multilateral resistance term. Why this specification is the canonical 2016+ structural-gravity workhorse.

Part IV

The empirical results: Spec B-prime and the regularization sweep

Chapter 12

Spec A baseline: replicating the McCallum-AvW finding

[STATUS: skeleton.] The 2010 panel with CEPII's `distw_harmonic` for both inter and intra. Replicates McCallum-AvW's $\sim 8\times$ multiplier. The clean baseline against which all sweeps are compared.

Chapter 13

Spec B-prime: the regularization sweep results

13.1 What Spec B-prime tests

Spec A (the baseline) uses CEPII's `distw_harmonic` for both intra and inter-national distance. Spec B-prime keeps CEPII's value for inter-national rows but replaces the intra-national distance with $d_{ii}^{\text{eff}}(\theta, d_{\min})$ computed numerically on the country polygon, for $\theta \in \{-3, -5, -7\}$ and $d_{\min} \in \{0.5, 1, 2, 5, 10, 20, 50, 100\}$.

Why hold inter-national distance fixed? The structural critique is specifically about the intra-national object. Inter-national distance is also a CES integral in principle, but the bilateral integral has support disjoint from the divergence at $r = 0$ (Volume 1 Chapter 11), so it converges at gravity-consistent θ . Swapping only the intra-national distance isolates the divergence-driven sensitivity.

13.2 The full 2010 sweep results

Three years (2010, 2015, 2020) $\times$ three structural θ 's $\times$ ten $d_{\min}$ cells $\times$ two baseline specs = 96 regressions. The headline 2010 results:

(lr)2-4 (lr)5-7 $d_{\min}$ (km)	$\hat{\gamma}$			$e^{\hat{\gamma}}$		
	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-1.155	-2.919	-3.360	$0.32\times$	$0.05\times$	$0.03\times$
1.0	-0.365	-2.054	-2.494	$0.69\times$	$0.13\times$	$0.08\times$
2.0	+0.308	-1.293	-1.733	$1.36\times$	$0.27\times$	$0.18\times$
5.0	+1.114	-0.378	-0.819	$3.05\times$	$0.68\times$	$0.44\times$
10	+1.601	+0.209	-0.231	$4.96\times$	$1.23\times$	$0.79\times$
20	+1.976	+0.705	+0.265	$7.21\times$	$2.02\times$	$1.30\times$
50	+2.336	+1.282	+0.842	$10.3\times$	$3.60\times$	$2.32\times$
100	+2.499	+1.580	+1.140	$12.2\times$	$4.86\times$	$3.13\times$
200	+2.593	+1.839	+1.399	$13.4\times$	$6.29\times$	$4.05\times$
500	+2.610	+2.031	+1.590	$13.6\times$	$7.62\times$	$4.91\times$

CEPII baseline at $\theta = -1$ (the conventional choice) gives $\hat{\gamma} = +2.132$, multiplier $8.43\times$. This sits at the high- $d_{\min}$ end of the sweep range.

13.3 Three observations

First, the home coefficient moves monotonically with $d_{\min}$ in every (θ, year) cell tested. As $d_{\min}$ shrinks (finer spatial resolution), $\hat{\gamma}$ becomes more negative. As $d_{\min}$ grows (coarser resolution, approaching the CEPII baseline grain), $\hat{\gamma}$ becomes more positive.

Second, the sign-flip threshold (where $\hat{\gamma}$ crosses zero) shifts cleanly with θ . At $\theta = -3$, the flip occurs between $d_{\min} = 1$ and $d_{\min} = 2$ km. At $\theta = -5$, between $d_{\min} = 5$ and $d_{\min} = 10$ km. At $\theta = -7$, between $d_{\min} = 10$ and $d_{\min} = 20$ km. More negative θ means the formula needs a larger $d_{\min}$ to produce a positive home coefficient.

Third, the conventional McCallum-Anderson-van Wincoop border-puzzle multiplier (the $5\times$ to $22\times$ range that the literature treats as “the answer”) sits inside the sweep range and is recovered at multiple $(\theta, d_{\min})$ combinations. For example, the $\sim 8\times$ multiplier appears at:

- $\theta = -3$, $d_{\min} = 20$ km ($7.21\times$)
- $\theta = -5$, $d_{\min} = 500$ km ($7.62\times$)
- $\theta = -7$, $d_{\min} > 500$ km (extrapolating)

The reduced-form $\theta = -1$ CEPII evaluation produces $8.43\times$ without committing to any structural θ . The literature’s border-puzzle multiplier is not separately identified from the regularization choice.

13.4 Cross-year robustness

The same sweep on year 2015 ($n_{\text{intra}} = 6$) and year 2020 ($n_{\text{intra}} = 8$) produces qualitatively similar patterns: home moves monotonically more negative as d_{min} shrinks. The *levels* differ across years because the small-N intra sample shifts (different countries pass the Wei filter in different years), but the within-year pattern is robust.

The empirical content of the divergence theorem (Volume 1 Theorem 11.1) is captured by the slope of $\hat{\gamma}$ versus $\log d_{\text{min}}$. The slope is steeper for more negative θ , consistent with the scaling argument in Volume 2 Chapter 2: at $\theta = -3$ the regularization-elasticity is $1/3$; at $\theta = -7$ it is $5/7$. The empirical sweep recovers exactly this prediction. The structural inconsistency in the intra-national distance object propagates into the home dummy through a mechanism that the closed-form math predicts.

Chapter 14

Cross-year stability and the small-N intra issue

[STATUS: skeleton.] Why n_{intra} varies (6 in 2015, 8 in 2020, 10 in 2010). Which countries appear in/out. The implication for cross-year level comparisons.

Chapter 15

The qualitative finding: home coefficient as a regularization artifact

[STATUS: skeleton.] The synthesis: a coefficient that the literature has treated as a structural border-effect parameter is in fact a function of an analyst-chosen discretization grain. The implications for the border-puzzle literature.

Part V

Forward path

Chapter 16

Anderson-Yotov multilateral resistance over time

[STATUS: skeleton.] The 2010 Anderson-Yotov paper on the multilateral-resistance term as a time-varying object. How the regularization issue interacts with the MR system.

Chapter 17

Sectoral and pair-year FE specifications

[STATUS: skeleton.] What changes when you move to sectoral trade data. Whether the regularization-sensitivity result survives the YPMM three-way FE specification.

Chapter 18

Welfare counterfactuals when d_{ii} does not exist

[STATUS: skeleton.] The Caliendo-Parro (2015) exact-hat methodology. Welfare effects of trade-policy changes computed without a single number for d_{ii} . The forward path for Paper 5b.

Chapter 19

What an honest gravity literature would do

[STATUS: skeleton.] The constructive proposal. Report border-puzzle multipliers as conditional on a regularization. Acknowledge the structural-versus-reduced-form θ distinction. Move to estimation strategies that do not require pinning d_{ii} to a single number.

Chapter A

Notation key (Volume 2 additions)

Symbol	Meaning	Units
N	Number of atoms in discretization	dimensionless
$d_{\min}$	Regularization cutoff	km
$\hat{d}_{ii}^{\text{eff}}$	Discretized CES expected pairwise distance	km
$\alpha(\theta)$	Regularization scaling exponent	dimensionless
ESRI : 54009	Mollweide equal-area projection	—
$X_{ij,t}$	Bilateral trade flow in year t	USD
$\eta_{i,t}$	Origin-year fixed effect	log units
$\mu_{j,t}$	Destination-year fixed effect	log units

Chapter B

Code listings index

The notebooks and modules referenced in Volume 2:

- `notebooks/19c_regularization_sensitivity.py` — Chapter 2
- `notebooks/18d_spec_b_prime_fixed.py` — Chapter 9
- `notebooks/18e_spec_b_prime_ppml.py` — PPML robustness check
- `src/paper5/region_raster.py` — Country polygon to RasterDist
- `src/paper5/ot_distance.py` — CES generalized-mean distance